

ONLINE PATH: Point-in-Time Closed-Loop Quantitative Inference & Counterfactual Verification (t -> t + H)

1. PIT Query & State

Task q_t , Data $\mathcal{D}_{\leq t}$

US / China A-Share / HK

Crypto Perps / Options

Strict Bitemporal Cutoff

2. RAG-Jev Progressive Router

$$s_{\text{route}} = \lambda s_{\text{lex}} + (1 - \lambda) \langle \phi(q_t), \phi(d_i) \rangle$$

129 Verified Skills (15 Plugins)

Stage-1 Card + Stage-2 Lazy Ref

4,039 Tokens (-98.91% vs. Full)

3. 64-KC Associative Readout

$$z_t = \text{TopK}_k(W_{\text{JEV}} \tilde{x}_t), k = 6$$

$$\hat{r}_t = (1 - \gamma(n)) \hat{y}_{\text{KC}} + \gamma(n) s_{\text{skill}}$$

64-KC Mushroom-Body + Prior Blend

Zero Backprop / Sparse WTA

4. 37 Guards & Settlement

$$g_j(\tau) = \mathbf{1}[\mathcal{M}(\tau) \equiv \mathcal{M}(\tilde{\tau}_j)]$$

AST + Future-Perturbation Probe

$$\text{Dopamine } G = \log(W_T/W_0)$$

Fail-Closed SHA-256 Receipt

Margin Telemetry

Gen-3 Cards (100% Top-1)

Gen-3 Woodbury+Fisher Op

OOS Return & Guard Trace

OFFLINE DUAL-LAYER FIN-RSI: Cryptographically Locked Co-Evolution of Discrete Skill Cards & Numeric Operators

LAYER A: Skill-Space RSI (Discrete Card Evolution)

[SKILL_HARNESS_LOCK]

A1. Margin Audit

$$\Delta(q) = s_{y^*} - \max_{j \neq y^*} s_j$$

36 Misses + 5 Thin (<15%)

A2. Card Mutation

+TRIGGER Anchors
+Sibling SKIP Repel

A3. 4-Gate Promote

Top-1: 66.7% -> 100%
Blind Holdout: 100%

Immutable Guard Invariant: Zero edits to evals/queries.jsonl, eval_triggers.py, eval_blind.py, validate.py

LAYER B: Operator-Space RSI (Readout Evolution)

[HARNESS_LOCK]

B1. ESS & SNR Probe

Sparse Cold-Start &
Crisis Drift Diagnosis

B2. Woodbury-Fisher

Rank-1 Σ_t^{-1} Update +
Fisher Ridge + VolGate

B3. Pareto Ledger

Sharpe: 1.16 -> 1.77
MaxDD: -10.5% -> -6.8%

Super-Additive Synergy with Layer A: +0.30 Sharpe (t = 6.23, p = 0.0034) & +0.0047 Rank IC (p = 6.5e-5)